

OBA 410/510 Sports Analytics

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COURSE OBJECTIVE

Analytical techniques and quantitative methods have been on the rise in many fields of business and are an integral part of managing sports-related enterprises. The sports domain involves diverse and rich data sources, enabling rigorous and data driven decision-making methodologies. These methodologies leverage mathematical analyses and statistical methods, including modern predictive-analytics tools. This course focuses on the theory and application of analytics in sports. The course also explores recent trends in sports analytics practice, with an emphasis on practical methodologies and their implementation. Data analytics methodologies apply to individual and team-based performance metrics, as well as other concerns (e.g., ticket sales, pricing, capacity) that impact overall organization profitability. The course's learning objectives, and the skillsets that the course helps develop, are therefore not limited to a particular function or sport.

Upon successful completion of this course, students will:

- Improve their overall problem solving and critical thinking ability.
- Gain an understanding of basic statistical concepts and their applications in the sports world.
- Obtain a broad survey of the methods used in sports data acquisition, processing, analysis, visualization and implementation.
- Become familiar with the full analytic life cycle: asking productive and relevant research questions, finding the right data, applying the appropriate tools, discovering insight, and clearly communicating results.

The course is intended for students interested in general management, operations, marketing or consulting, and it leverages students' core-course data management skills relating to spreadsheets and databases. Students will work on integrating those data management skills with statistics-based data-analytics methods, and thus be able to make rigorous data-driven decisions, as well as effectively communicate the rationale supporting those decisions.

COURSE MATERIALS**Textbook (Optional but Recommended):**

Analytic Methods in Sports by Thomas A. Severini. 2nd edition, 2020. ISBN 9780367469382.

Mathletics: How Gamblers, Managers, and Sports Enthusiasts Use Mathematics in Baseball, Basketball, and Football by Wayne Winston Princeton University Press, 2009.

[Journal of Sports Analytics](#)

[MIT Sloan Sports Analytics Conference](#)

Software:

This course will use MS Excel and Python. You are also recommended to install Anaconda, an open-source coding platform with a built-in Python compiler:

- For Windows: <https://docs.anaconda.com/anaconda/install/windows/>
- For MacOS: <https://docs.anaconda.com/anaconda/install/mac-os/>

TENTATIVE SCHEDULE* Below is a tentative schedule of this class.

Topics

- 1 Intro. to Sports Analytics (Sports Performance and Data)
- 2 Data Visualization
 - [NYT Learning Network - What's Going On in This Graph?](#)
 - **Discussion** - Groeger, L. "[Design Principles for News Apps & Graphics.](#)" *ProPublica*, May 30, 2013.
 - [The Athletic's Match Dashboard](#)
 - [Radar Plots](#)
- 3 Sports Statistics, Probability Theory
- 4 Predictive Models in Sports
 - Binary Outcome and Regression
 - Logistic Regression
 - LPM and Logit Model
 - Ordered Logit Regression
 - Basics of Forecasting
- 5 Data Science for Sports
 - Analytics on the Field of Play
 - Player Evaluations
 - Strategic Decision Making
 - Ticketing, Fan Engagement
 - Evaluating Sports Sponsorships
- 6 Collecting and Working with Sports Data
- 7 Data Visualization - Unlock the Potential of Generative AI in Sports
 - **Understanding Training Data:** Identify biases and anomalies in data, and ensure that the data accurately represents real-world problems for generative AI.
 - **Performance Evaluation:** Visualize generative AI outputs to detect errors or artifacts, and enhance the training process and make adjustments to the model architecture as needed.
 - **Communication of Results:** Craft informative and visually compelling visualizations, and assist others in comprehending the insights derived from generative AI models.
 - **Debugging:** Visualize outputs at various training stages, and identify and rectify errors in both the training data and model architecture.
 - **Interpretation of Predictions:** Utilize visualizations to discern key features influencing predictions.
 - **Model Development:** Employ data visualization to investigate relationships between different features, and streamline the development of new generative AI models.